

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} M_x & 0 & 0 \end{bmatrix}$$

$$M_x = \mathbb{M}_x^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & 0 & M_{xz} \\ 0 & 0 & 0 \\ M_{xz} & 0 & 0 \end{bmatrix}$$

$$M_{xz} = \mathbb{M}_{xz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & M_{xz} \\ 0 & 0 & 0 \\ -M_{xz} & 0 & 0 \end{bmatrix}$$

$$M_{xz} = -\mathbb{T}_y^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & 0 & 0 \\ M_{yyx} & 0 & 0 \\ M_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M_{zzx} \\ 0 & M_{yyx} & 0 \end{bmatrix}$$

$$M_{xxx} = 3\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)}$$

$$M_{yyx} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\beta(1)}$$

$$M_{zzx} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\beta(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ M_{yyx} & 0 & 0 \\ M_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{M_{zzx}}{2} \\ 0 & -\frac{M_{yyx}}{2} & 0 \end{bmatrix}$$

$$M_{yyx} = 2\mathbb{M}_x^{(2)} + 2\mathbb{T}_{yz}^{(1)}$$

$$M_{zzx} = 2\mathbb{M}_x^{(2)} - 2\mathbb{T}_{yz}^{(1)}$$

$$\frac{M_{zzx}}{2} = \mathbb{M}_x^{(2)} - \mathbb{T}_{yz}^{(1)}$$

$$\frac{M_{yyx}}{2} = \mathbb{M}_x^{(2)} + \mathbb{T}_{yz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{zzz} \\ 0 & M_{xyy} & 0 \end{bmatrix}$$

$$M_{zzz} = \mathbb{M}_x^{(3)} + \mathbb{T}_{yz}^{(2)}$$

$$M_{xyy} = -\mathbb{M}_x^{(3)} + \mathbb{T}_{yz}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & M_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ M_{xxzx} & M_{yyzx} & M_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = -\mathbb{M}_{4y}^{\alpha(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_y^{\beta(1)}$$

$$M_{yyzx} = \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_y^{\beta(1)}$$

$$M_{zzzx} = \mathbb{M}_{4y}^{\alpha(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_y^{\beta(1)}$$

$$M_{yzxy} = \mathbb{M}_{xz}^{(2)} + 2\mathbb{M}_y^{\beta(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & M_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ -M_{xxzx} & -M_{yyzx} & -M_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = \mathbb{M}_{xz}^{(3)} - 2\mathbb{T}_y^{(1)} + 2\mathbb{T}_y^{\alpha(1)} + 2\mathbb{T}_y^{\beta(1)}$$

$$M_{yyzx} = \mathbb{M}_{xz}^{(3)} - 4\mathbb{T}_y^{\beta(1)}$$

$$M_{zzzx} = \mathbb{M}_{xz}^{(3)} + 2\mathbb{T}_y^{(1)} - 2\mathbb{T}_y^{\alpha(1)} + 2\mathbb{T}_y^{\beta(1)}$$

$$M_{yzxy} = \mathbb{T}_y^{(1)} + 4\mathbb{T}_y^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{yzxy} = 2\mathbb{M}_{xz}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ -M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{yzxy} = \mathbb{T}_y^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxzx} & 0 \\ 0 & M_{yyzx} & 0 \\ 0 & M_{zzzx} & 0 \\ 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ M_{xyyz} & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = 2\mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(3)} - \mathbb{T}_y^{\alpha(2)} - \mathbb{T}_y^{\beta(2)}$$

$$M_{yyzx} = \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{(4)} + 2\mathbb{T}_y^{\alpha(2)}$$

$$M_{zzzx} = -2\mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(3)} - \mathbb{T}_y^{\alpha(2)} + \mathbb{T}_y^{\beta(2)}$$

$$M_{yzxy} = \mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(4)} - \mathbb{T}_y^{\alpha(2)} + \mathbb{T}_y^{\beta(2)}$$

$$M_{xyyz} = -\mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(4)} - \mathbb{T}_y^{\alpha(2)} - \mathbb{T}_y^{\beta(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ M_{zzxx} & M_{zxyy} & M_{zxzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{xyyz} & 0 & 0 \end{bmatrix}$$

$$M_{yzxy} = -\mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)}$$

$$M_{zzxx} = 2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(5)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)}$$

$$M_{zxyy} = \mathbb{T}_y^{(5)} + 2\mathbb{T}_y^{(6)} + 2\mathbb{T}_y^{\alpha(3)}$$

$$M_{zzzz} = -2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(5)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)}$$

$$M_{xyyz} = \mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 & 0 & M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & M_{yyzx} & 0 & 0 & M_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & M_{zzzx} & 0 & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ M_{zzxx} & M_{zxyy} & M_{zxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{xyyz} & 0 & 0 & M_{xyzy} & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = -\mathbb{M}_{4y}^{\alpha(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)} - \mathbb{M}_y^{\beta(1)} - 2\mathbb{T}_y^{(1)} + \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} - \mathbb{T}_y^{\beta(2)}$$

$$M_{xxxz} = -\mathbb{M}_{4y}^{\alpha(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} - \mathbb{M}_y^{\beta(1)} - 2\mathbb{T}_y^{(1)} - \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} + \mathbb{T}_y^{\beta(2)}$$

$$M_{yyzx} = \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{(4)} + 2\mathbb{T}_y^{\alpha(2)} - 4\mathbb{T}_y^{\beta(1)}$$

$$M_{yyxz} = \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{T}_y^{(3)} - 2\mathbb{T}_y^{(4)} - 2\mathbb{T}_y^{\alpha(2)} - 4\mathbb{T}_y^{\beta(1)}$$

$$M_{zzzz} = \mathbb{M}_{4y}^{\alpha(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} - \mathbb{M}_y^{\beta(1)} + 2\mathbb{T}_y^{(1)} + \mathbb{T}_y^{(3)} - 2\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} + \mathbb{T}_y^{\beta(2)}$$

$$M_{zzxz} = \mathbb{M}_{4y}^{\alpha(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)} - \mathbb{M}_y^{\beta(1)} + 2\mathbb{T}_y^{(1)} - \mathbb{T}_y^{(3)} - 2\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} - \mathbb{T}_y^{\beta(2)}$$

$$M_{yzxy} = \mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(5)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_y^{(1)} + \mathbb{T}_y^{(4)} + 4\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} + \mathbb{T}_y^{\beta(2)}$$

$$M_{yzyx} = \mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(5)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_y^{(1)} - \mathbb{T}_y^{(4)} + 4\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} - \mathbb{T}_y^{\beta(2)}$$

$$M_{zzxx} = -\mathbb{M}_{4y}^{\alpha(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)} - \mathbb{M}_y^{\beta(1)} + 2\mathbb{T}_y^{(1)} - 2\mathbb{T}_y^{\alpha(1)} - 2\mathbb{T}_y^{\beta(1)}$$

$$M_{zxyy} = \mathbb{M}_{xz}^{(1)} - \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_y^{\beta(1)} + 4\mathbb{T}_y^{\beta(1)}$$

$$M_{zzzz} = \mathbb{M}_{4y}^{\alpha(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)} - \mathbb{M}_y^{\beta(1)} - 2\mathbb{T}_y^{(1)} + 2\mathbb{T}_y^{\alpha(1)} - 2\mathbb{T}_y^{\beta(1)}$$

$$M_{xyyz} = \mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(5)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{T}_y^{(1)} + \mathbb{T}_y^{(4)} - 4\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} - \mathbb{T}_y^{\beta(2)}$$

$$M_{xyzy} = \mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(5)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{T}_y^{(1)} - \mathbb{T}_y^{(4)} - 4\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} + \mathbb{T}_y^{\beta(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzyx} \\ M_{zzxx} & M_{zxyy} & M_{zxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{xyyz} & 0 & 0 & M_{xyzy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yzxy} &= 2\mathbb{M}_{xz}^{(4)} - \mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(2)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)} \\
M_{yzyx} &= -2\mathbb{M}_{xz}^{(4)} - \mathbb{M}_{xz}^{(6)} - \mathbb{T}_y^{(2)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)} \\
M_{zzxx} &= 2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(5)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)} \\
M_{zxyy} &= \mathbb{T}_y^{(5)} + 2\mathbb{T}_y^{(6)} + 2\mathbb{T}_y^{\alpha(3)} \\
M_{zzzz} &= -2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(5)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)} \\
M_{xyyz} &= 2\mathbb{M}_{xz}^{(4)} + \mathbb{M}_{xz}^{(6)} - \mathbb{T}_y^{(2)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)} \\
M_{xyzy} &= -2\mathbb{M}_{xz}^{(4)} + \mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(2)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
0 & 0 & M_{xxxx} \\
0 & 0 & 0 \\
M_{zzzz} & 0 & 0 \\
M_{yyyz} & 0 & 0 \\
0 & 0 & M_{zzxz} \\
0 & 0 & 0 \\
0 & 0 & 0 \\
M_{zzxx} & 0 & 0 \\
0 & 0 & M_{xyyz} \\
0 & M_{xyyz} + \frac{M_{zzxz}}{2} - \frac{M_{zzxx}}{2} & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= -\mathbb{M}_{4y}^{\alpha(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} - \mathbb{M}_y^{\beta(1)} \\
M_{zzzx} &= \mathbb{M}_{4y}^{\alpha(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} - \mathbb{M}_y^{\beta(1)} \\
M_{yyzx} &= \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_y^{\beta(1)} \\
M_{zzxz} &= \mathbb{M}_{4y}^{\alpha(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)} - \mathbb{M}_y^{\beta(1)} \\
M_{zzxx} &= -\mathbb{M}_{4y}^{\alpha(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)} - \mathbb{M}_y^{\beta(1)} \\
M_{xyyz} &= \mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(5)} + 2\mathbb{M}_y^{\beta(1)}
\end{aligned}$$